

Business Questions

Sprint 4

Q16: How long does it take a student (on average) to complete a reservation from opening the app until final confirmation?

userId	Average of seconds_to_search
18afd5b5-08bf-45d0-82cd-e671a8fa8fef	201.59
9556394d-6d16-42af-8e91-d962c23dceee	201.59
20608103-cb27-42e5-9b87-b949664042ad	187.20
28c925f0-75b8-461f-b339-0f5cf1c114e8	187.19
0d3fb66b-78b2-4df3-9bcd-e2b2df8e0fa8	172.80
4e7482ec-c60a-4fc9-a9a4-e1515d308682	158.40
6cb76d8b-9bfc-4748-acaa-6b2edaf626f9	158.39
Total	158.39

Average of time (sec)

158.39

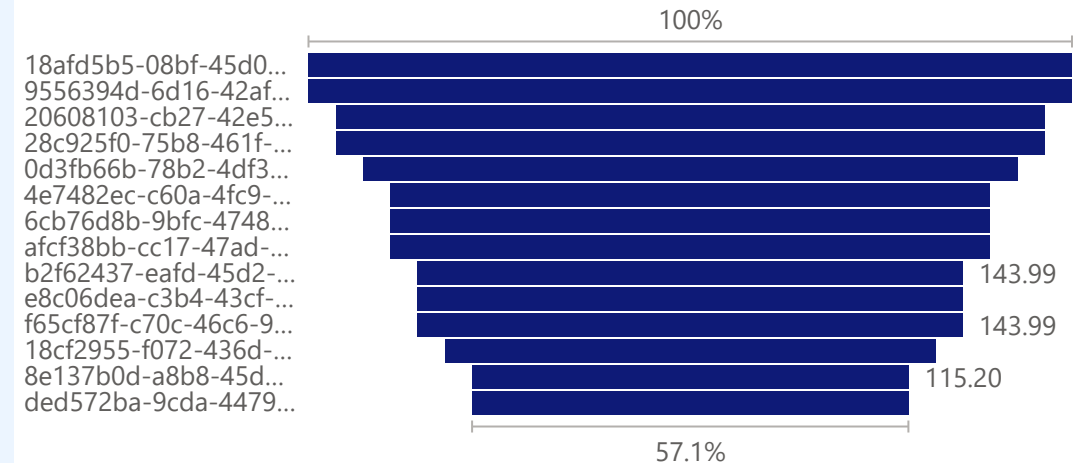
Range Days

t_first_search

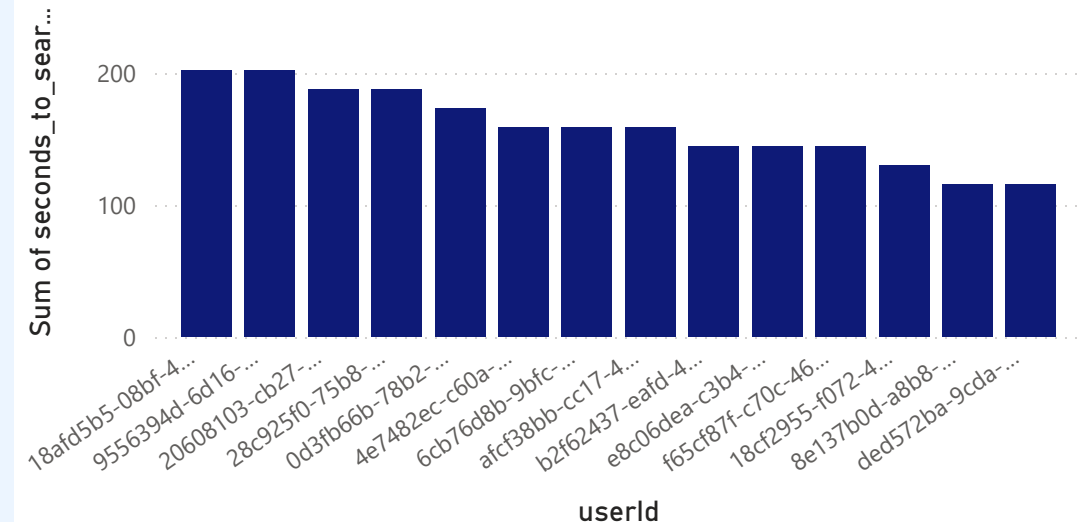
9/20/2025 9/26/2025

Range slider showing dates from 9/20/2025 to 9/26/2025.

Duration to make to make reservations



Duration to make to make reservations



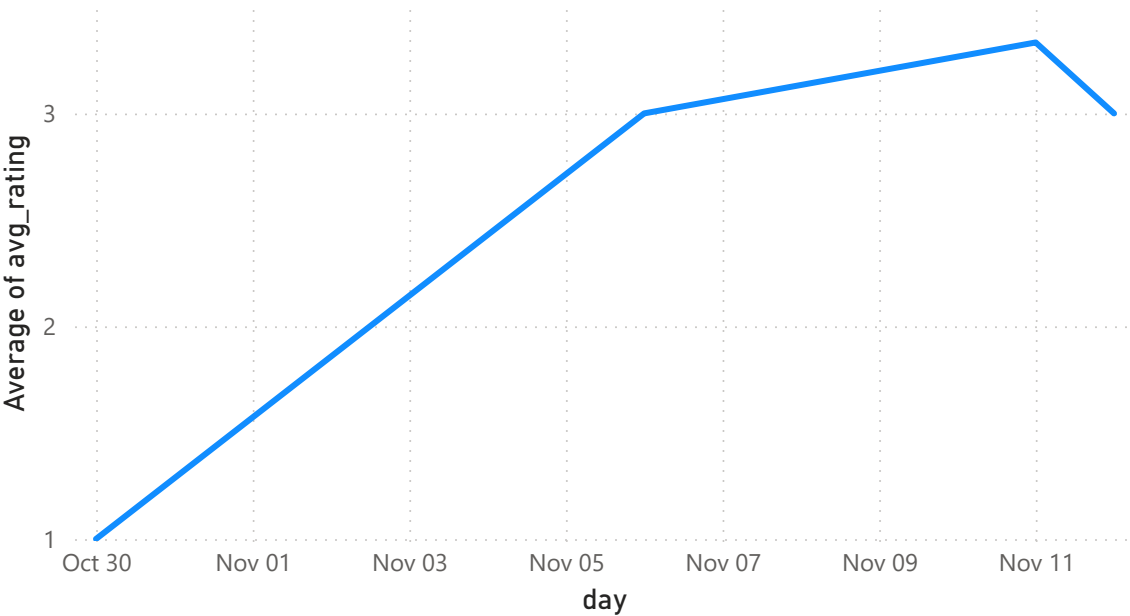
Q17: How does each space’s average rating evolve over time (cleanliness, safety, comfort)?

Number of reviews

6

space_id	Sum of n_reviews	day	avg_rating
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	1	Wednesday, November 12, 2025	3.00
0429d5f7-bbc0-4ca7-b21a-81b5f5379685	1	Tuesday, November 11, 2025	2.00
50161560-1379-4dce-b17f-3ad3e9c99768	1	Tuesday, November 11, 2025	4.00
589ec857-3cd6-4aa7-aa47-757739dfd5fa	1	Thursday, November 06, 2025	3.00
5d653052-eb6a-4054-bbe7-14c6479b5d29	1	Thursday, October 30, 2025	1.00
6922acef-a9c4-48f9-a003-554a1a18b6db	1	Tuesday, November 11, 2025	4.00
Total	6		

Average rating by day



Q18: What is the probability of a no-show depending on user segment (semester, major)?

major	semester	total_bookings	p_no_show_proxy
(Unknown)	-1	9	0.11
Administración	2	47	0.02
Administración	9	12	0.75

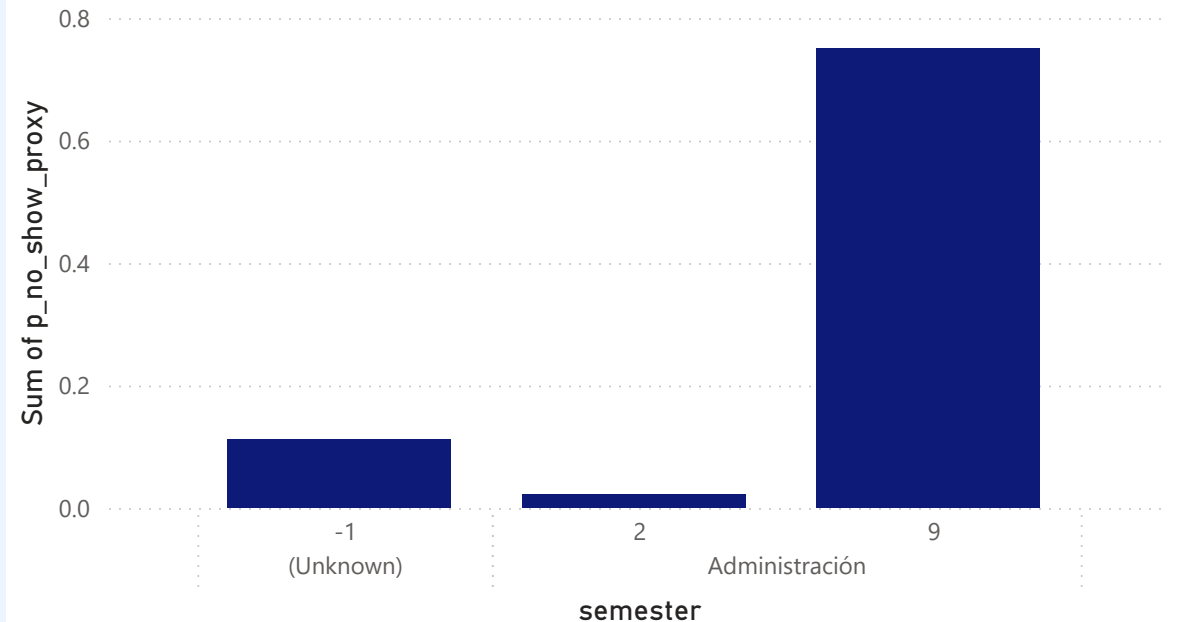
Number of semesters

3

Average of probability

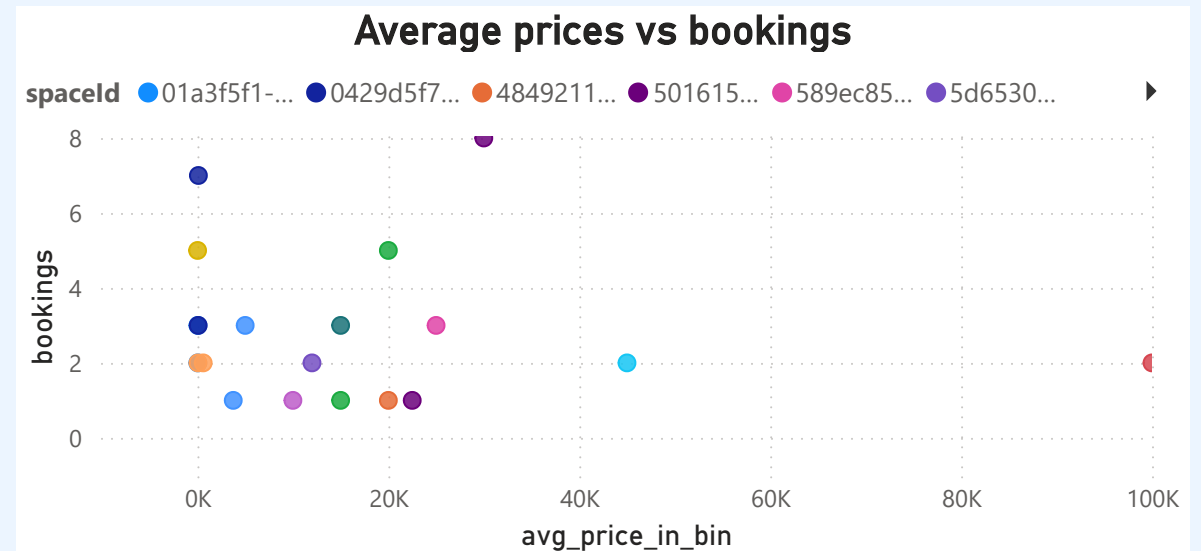
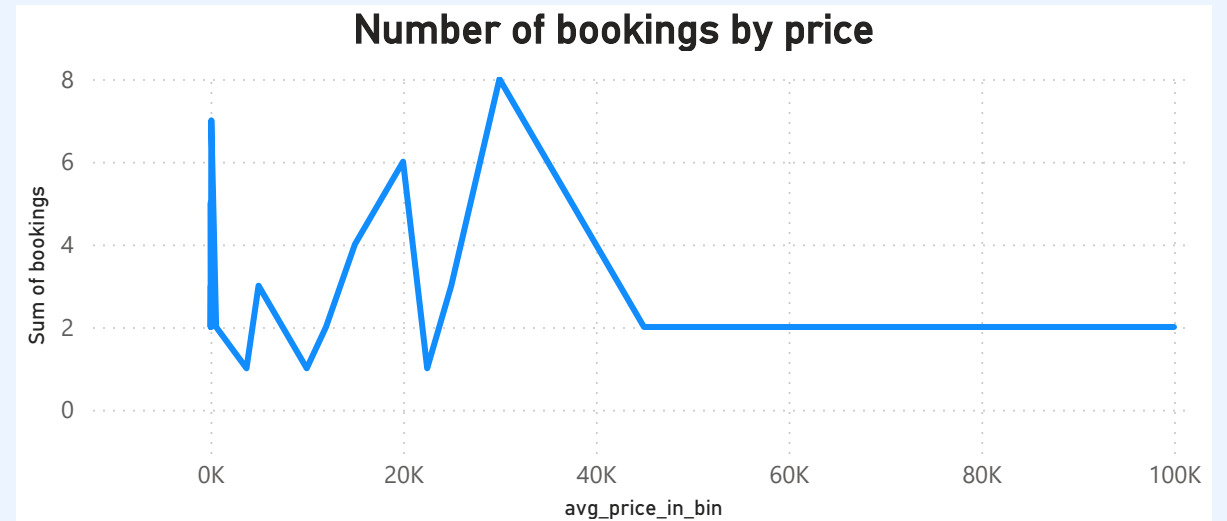
0.29

probability of a no-show depending on user segment



Q19: How does a change in hourly price ($\pm 10\text{--}20\%$) affect demand for each space type?

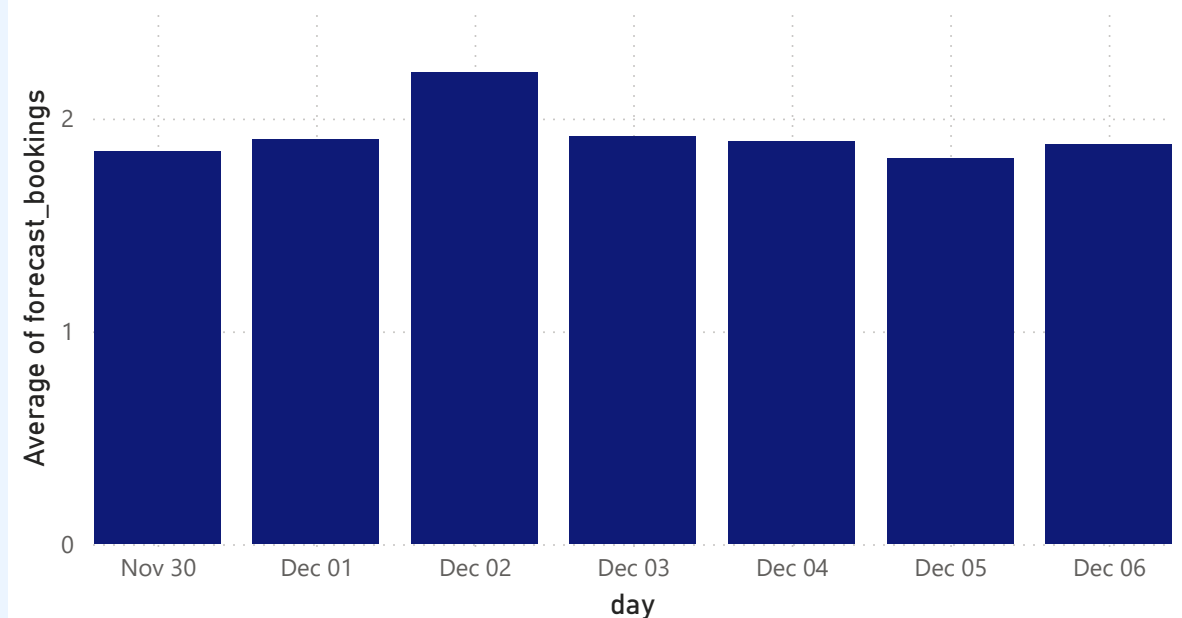
avg_price_in_bin	price_bin_10s	bookings
15.00	1	5
37.50	3	2
50.00	5	3
53.09	5	2
74.25	7	3
99.00	9	7
600.00	60	2
3,750.00	375	1
5,000.00	500	3
10,000.00	1000	1
12,000.00	1200	2
15,000.00	1500	1
15,000.00	1500	3
20,000.00	2000	1
20,000.00	2000	5



Q22: Based on past data, what is the likelihood that a student will extend their booking?

space_id	day	forecast_bookings
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Sunday, November 30, 2025	2.50
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Monday, December 01, 2025	2.50
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Tuesday, December 02, 2025	4.00
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Wednesday, December 03, 2025	2.50
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Thursday, December 04, 2025	2.50
01a3f5f1-7fc7-4e77-b05e-3b4cb63e3afb	Friday, December 05, 2025	1.00
01a3f5f1-7fc7-4e77-	Saturday,	2.50

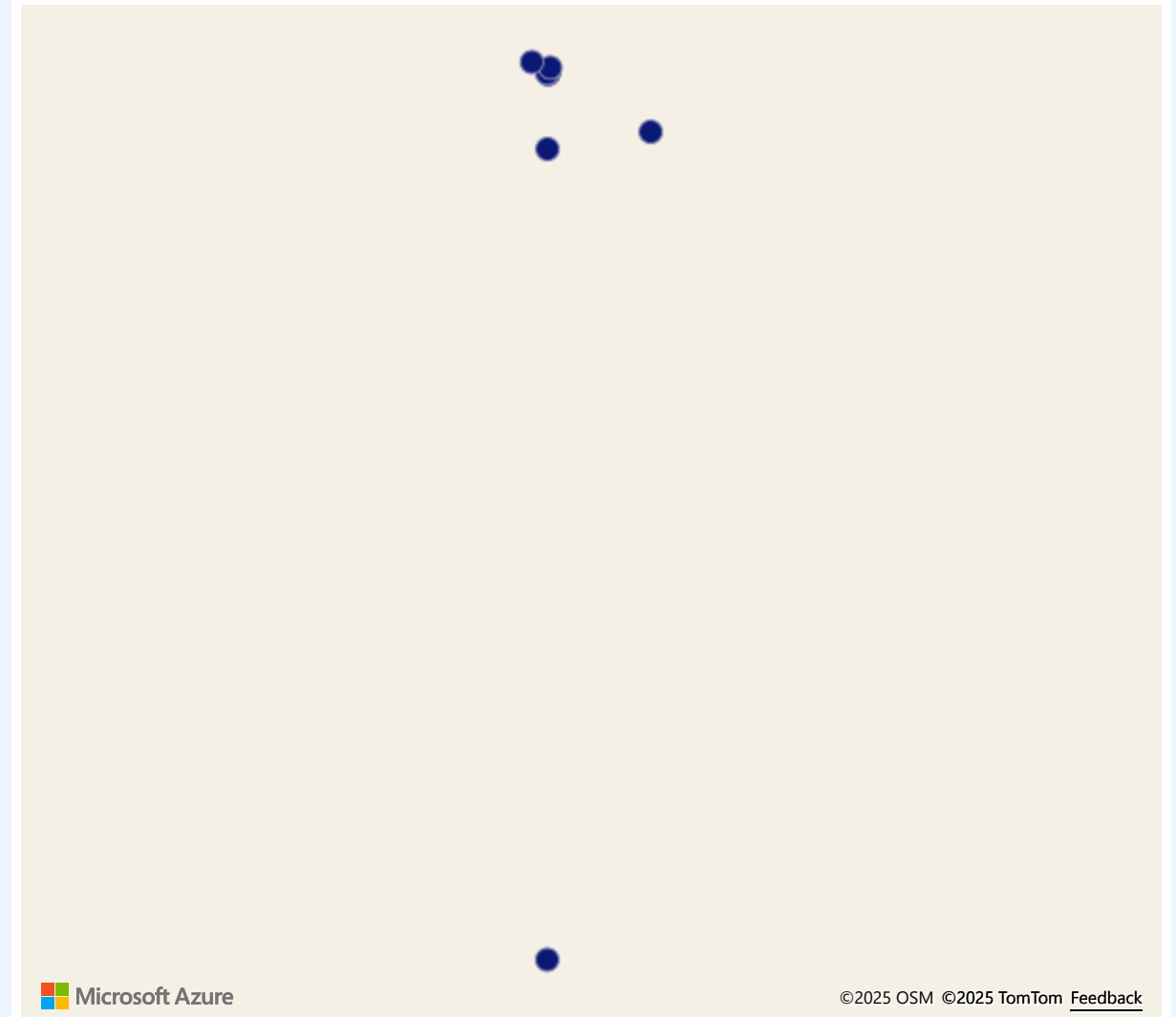
Valor esperado de reservas por dias futuros



Q23: Are there natural clusters of high-demand areas near campus where adding new rooms would increase availability?

space_id	title	lon	lat
c3c53cb2-51c7-4a5b-87cc-5ebcc801d222	jdidsnj	-73.00	4.00
af792b21-750b-4ca6-a0e5-299efdac9a8e	Sala de Estudio Silenciosa - Edificio ML	-74.04	4.67
bb6e54f2-05bb-424f-b289-e4c92356a126	Área de Coworking - Edificio W	-74.06	4.60
5d653052-eb6a-4054-bbe7-14c6479b5d29	Espacio de Lectura - Biblioteca	-74.06	4.60
589ec857-3cd6-4aa7-aa47-757739dfd5fa	Sala de Reuniones Grupales - Edificio SD	-74.07	4.60
6922acef-a9c4-48f9-a003-554a1a18b6db	Sala de Estudio 101	-74.07	4.60
c5e39f0c-5747-457a-	Habitación con	-74.07	4.60

Demand per space



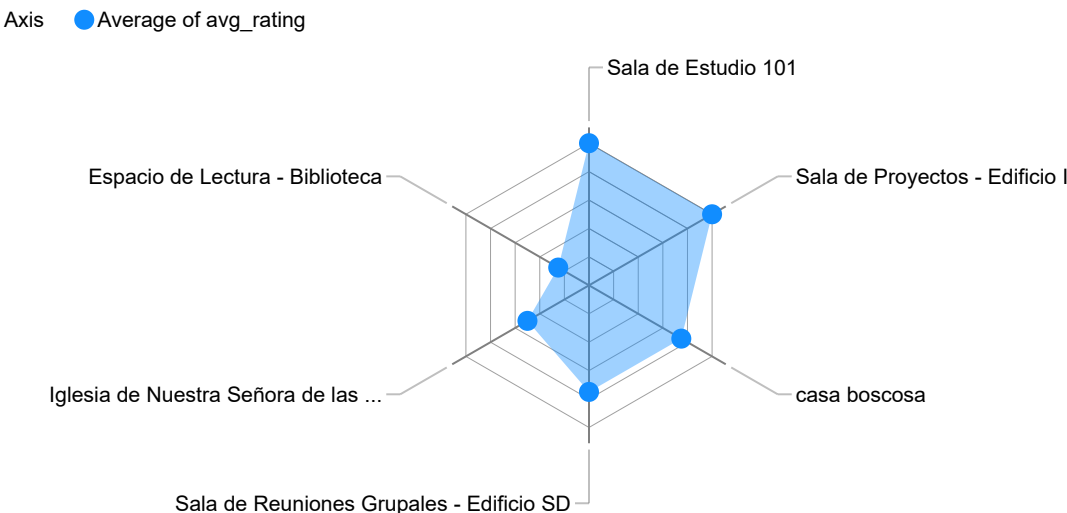
Business Questions

Sprint 3

Q5: How satisfied are students with the reserved spaces in terms of cleanliness, safety, and comfort?

space_name	min_rating	max_rating	avg_rating
casa boscosa	3	3	3.00
Espacio de Lectura - Biblioteca	1	1	1.00
Iglesia de Nuestra Señora de las Aguas	2	2	2.00
Sala de Estudio 101	4	4	4.00
Sala de Proyectos - Edificio I	4	4	4.00
Sala de Reuniones Grupales - Edificio SD	3	3	3.00

Score per location



Filtro sitios

- ☒ Select all
- ☒ (Blank)
- ☒ casa boscosa
- ☒ Espacio de Lectura - Biblioteca
- ☒ Iglesia de Nuestra Señora de las Aguas
- ☒ Sala de Estudio 101
- ☒ Sala de Proyectos - Edificio I

Q9: How often do students use the comments and ratings section, and how does this influence bookings?

Average number of reservations per space

4.86

Average number of ratings per space

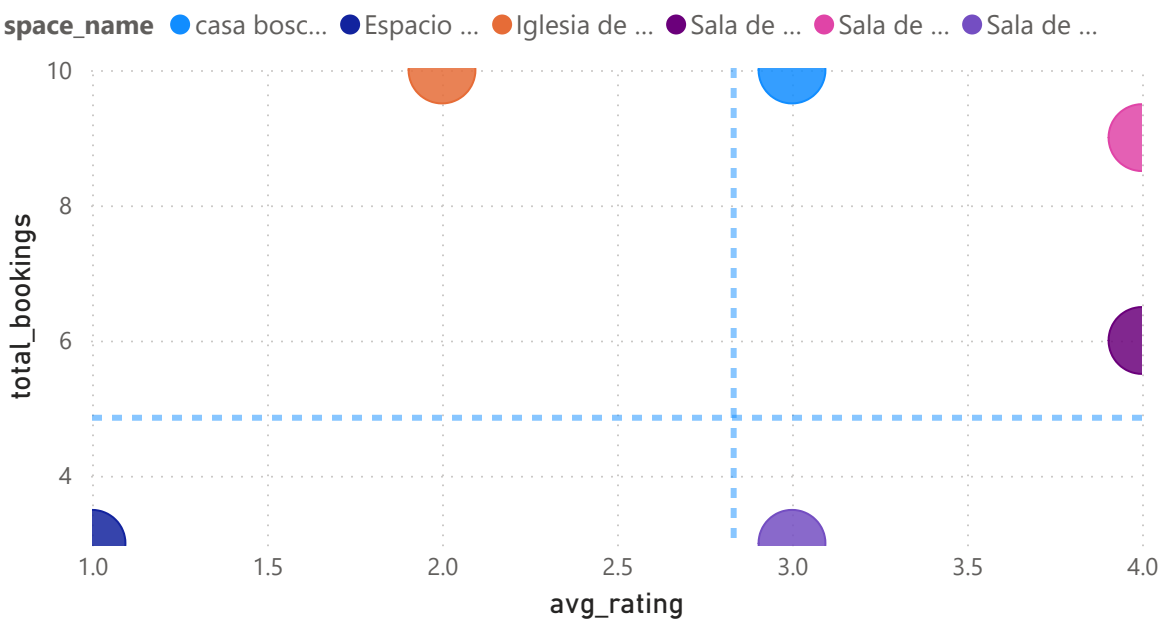
2.83

Average number of reviews per space

1.00

space_name	Sum of total_bookings	Sum of review_count	Sum of avg_rating
Área de Coworking - Edificio W	7		
casa boscosa	10	1	3.00
casa de juan	5		
casa de juanita	2		
Espacio de Lectura - Biblioteca	3	1	1.00
Eternal	1		
Habitacion City U 1	1		
Habitación con vista al campus	2		
Habitacion san jose	5		
Iglesia de Nuestra Señora de las Aguas	10	1	2.00
Total	68	6	17.00

Number of cancellations per day



Q10: Based on past data, what is the likelihood that a student will extend their booking?

Sum of total_bookings	Sum of extended_bookings	extension_rate
68	0	0.00
68	0	

Probability of extension

0.00

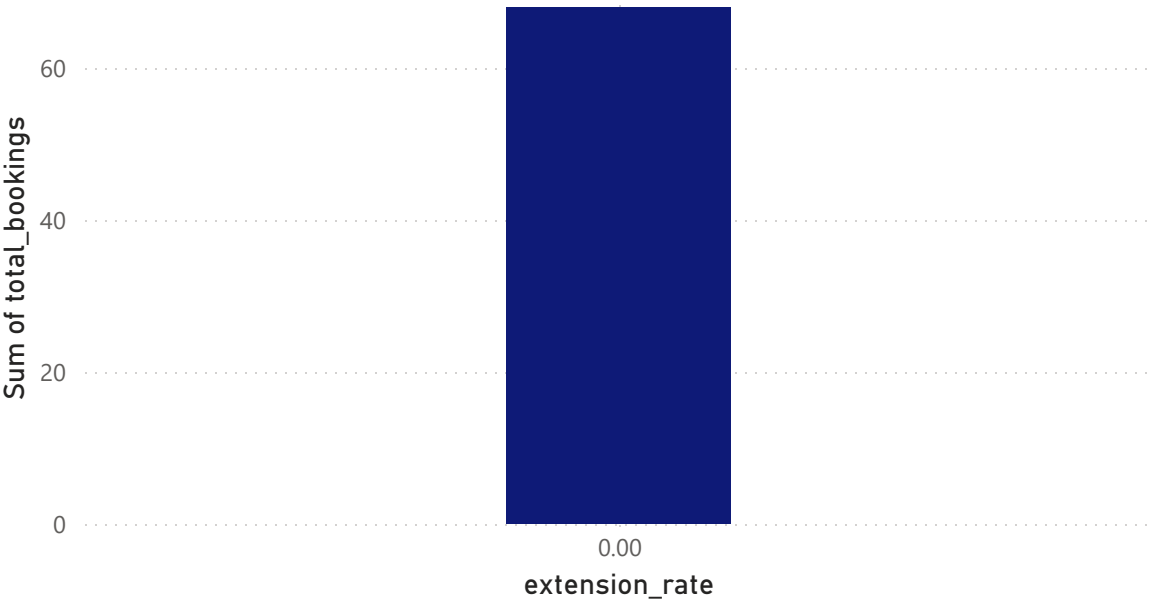
Number of bookings

68

Number of extended bookings

0

Extension frequency



Sprint 3

Business Question 12

Q12: Which campus areas or buildings show the highest demand for hourly rentals?

space_name	avg_price	total_bookings
Área de Coworking - Edificio W	20,000.00	6
casa boscosa	100.00	5
casa de juan	3,000.00	4
casa de juanita	100,000.00	2
Espacio de Lectura - Biblioteca	12,000.00	2
Eternal	10,000.00	1

Demand per space

space_na... Iglesia de Nuestr... Sala de Pr... ▶

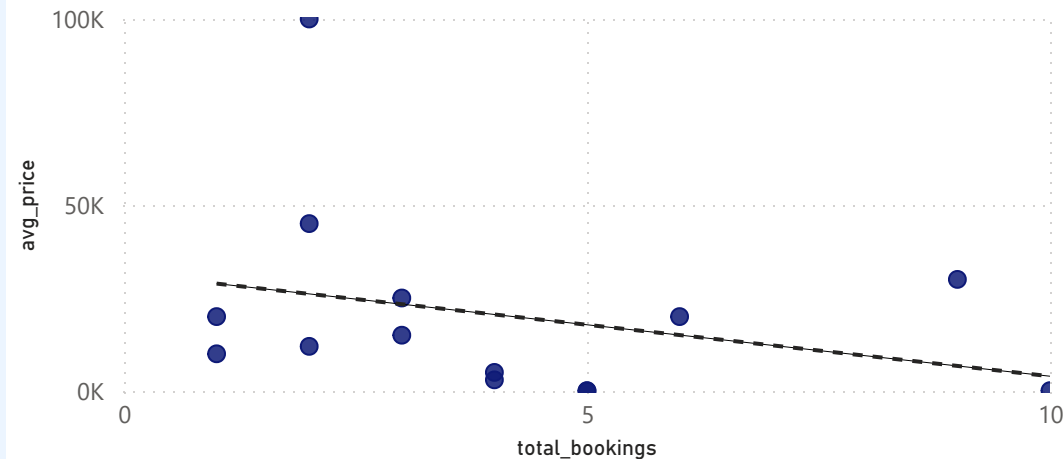


Price per space

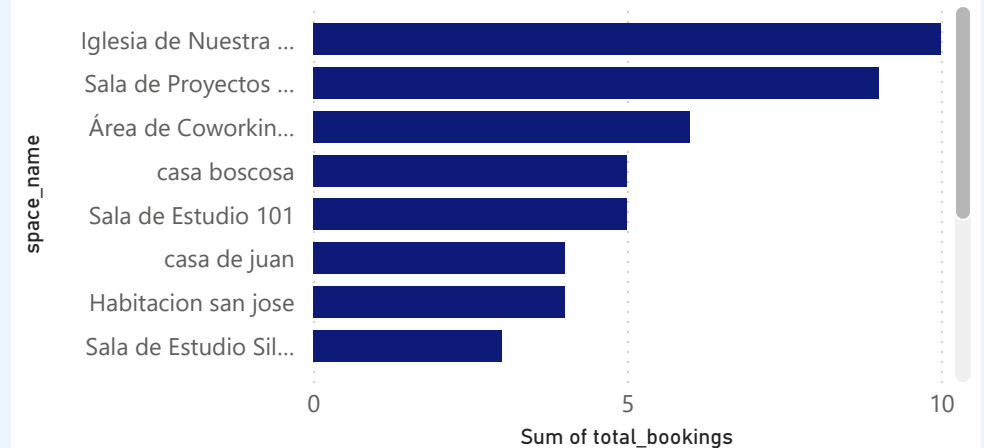
space_na... casa de juanita Habitación con... ▶



Relationship between price and demand

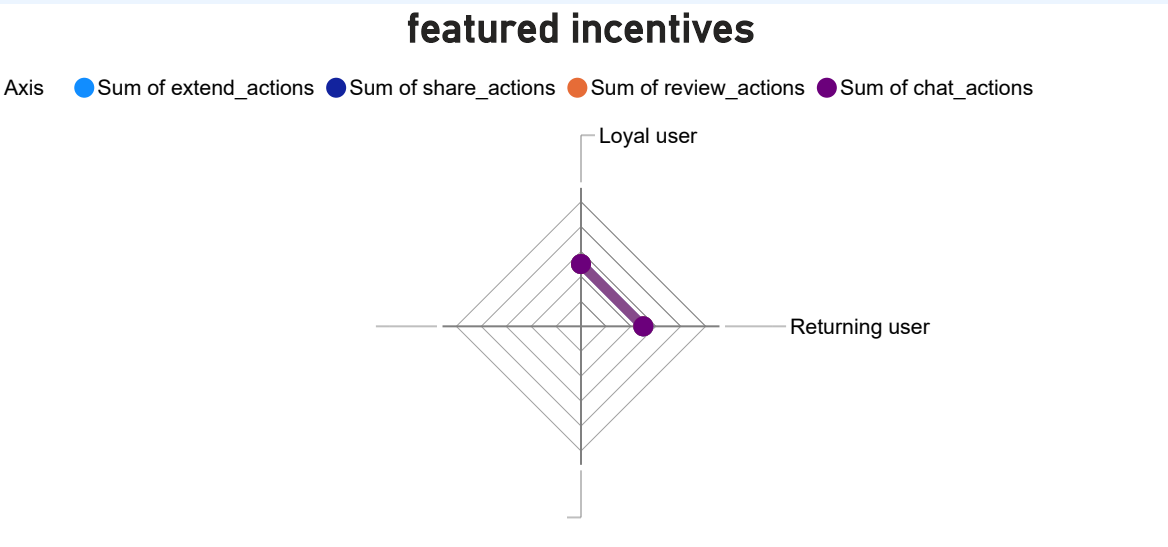
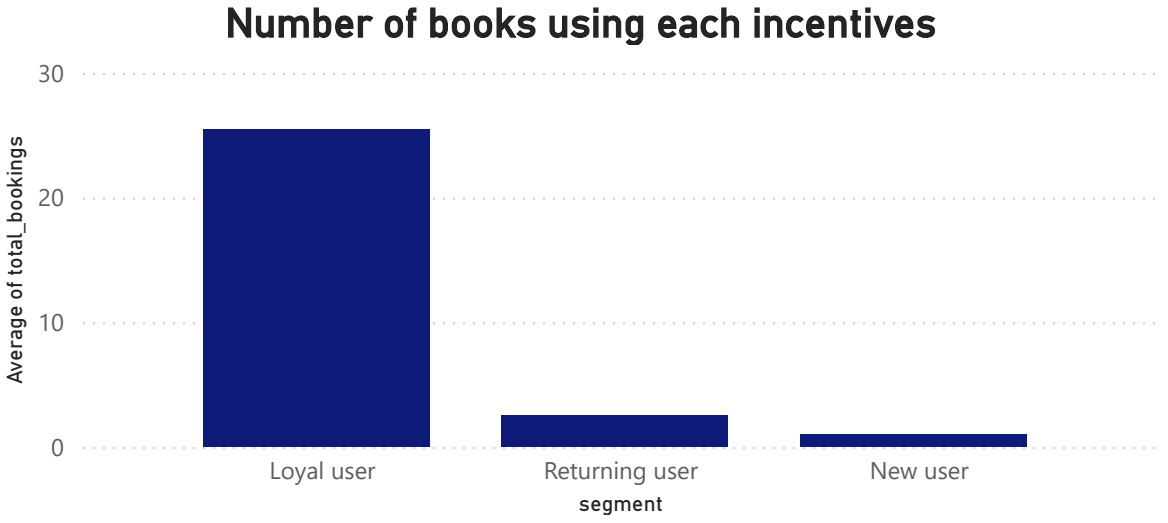


Ranking on demand



Q14: What incentives (discounts, loyalty points, group bookings) would most effectively increase repeat usage?

segment	review_actions	extend_actions	chat_actions	Sum of total_bookings	share_actions
Loyal user				5	
Loyal user	0	0	0	46	
New user				1	
Returning user				2	
Returning user	0	0	0	3	
Total				57	



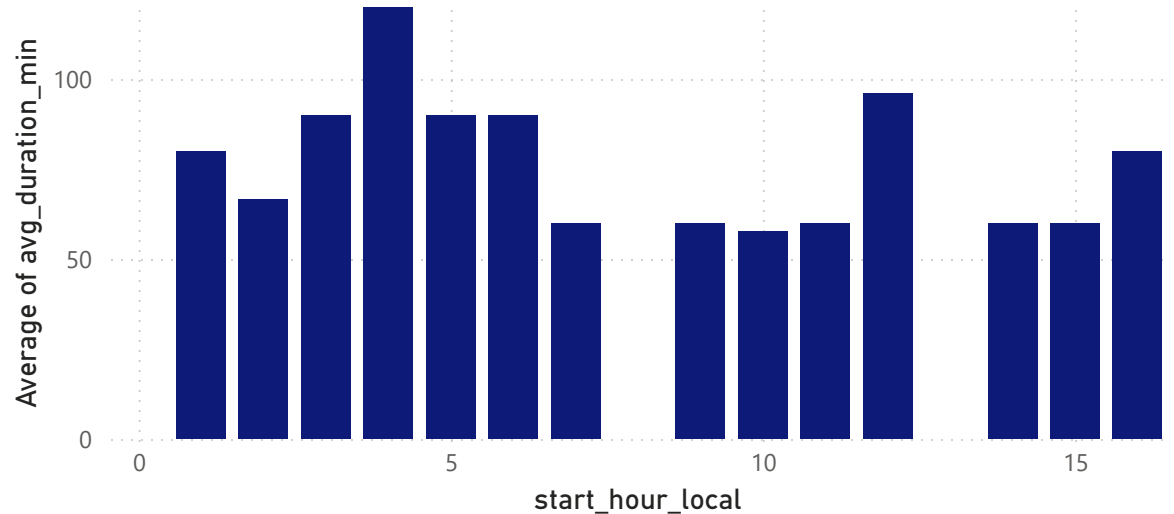
Q15: At what hours of the day do students book spaces the most, and how long do they typically stay?

start_hour_local	Average of avg_duration_min	Sum of total_reservations
1	80.00	3
2	66.67	9
3	90.00	4
4	120.00	3
5	90.00	2
6	90.00	4
7	60.00	1
9	60.00	3
10	57.67	3
Total	76.45	57

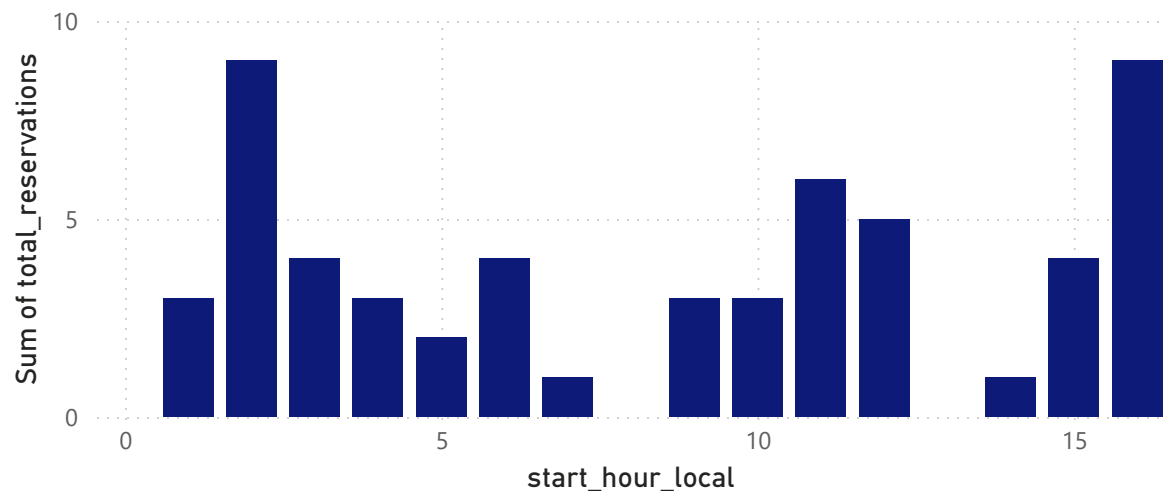
Average duration of books (mins)

76.45

Average time of duration per start hour



Number of books per start hour



Business Questions

Sprint 2

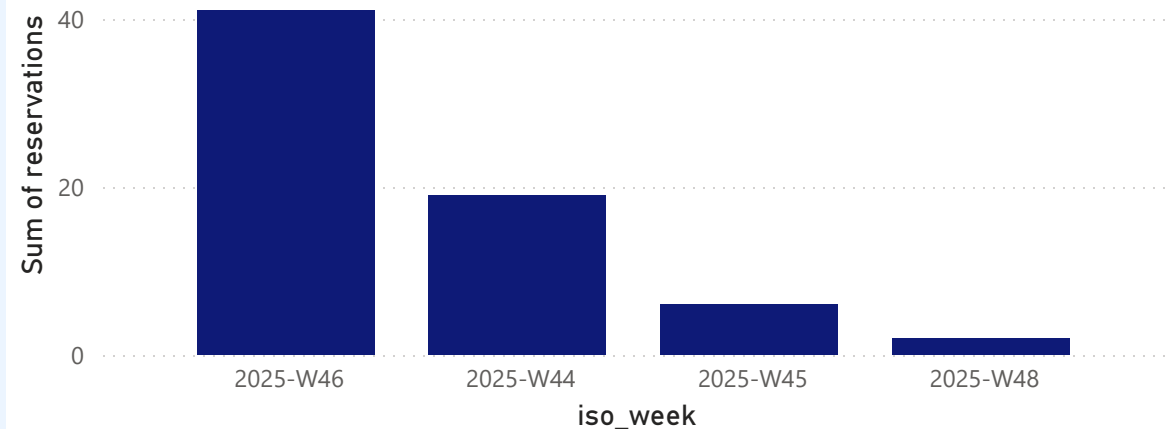
Q2: How many reservations are made per week, and what is the average booking duration?

343.64

Sum of avg_duration_min

iso_week	reservations	Sum of avg_duration_min
2025-W44	19	82.11
2025-W45	6	130.00
2025-W46	41	71.54
2025-W48	2	60.00
Total		343.64

Books per week

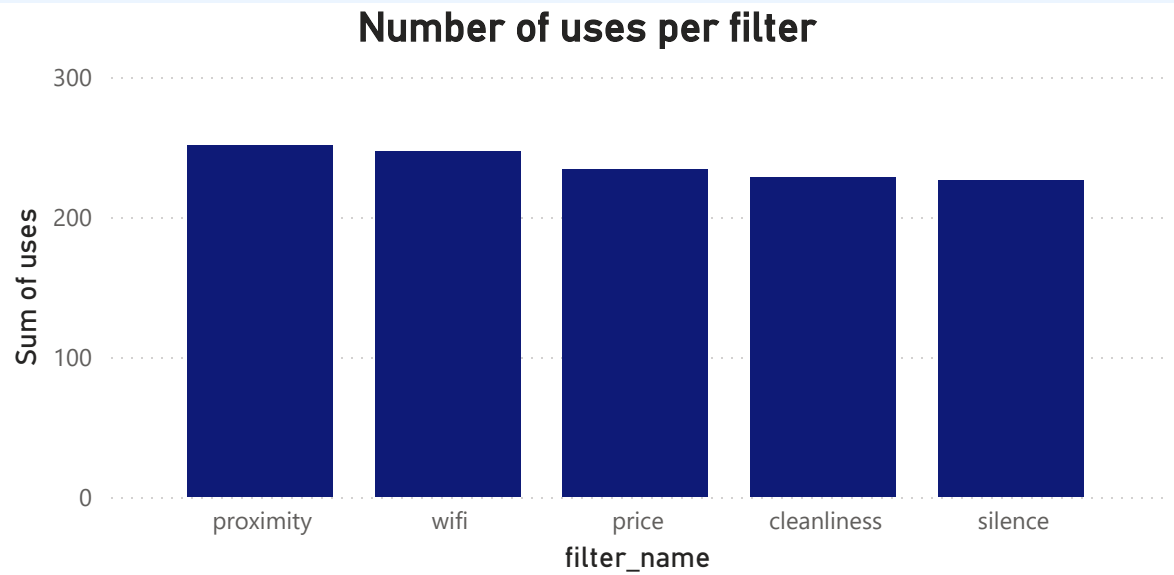


Q3: Which filters (silence, price, proximity, Wi-Fi, cleanliness) are most used by students when searching for spaces?

237.20

Average of uses

filter_name	Sum of uses
cleanliness	228
price	234
proximity	251
silence	226
wifi	247
Total	1186



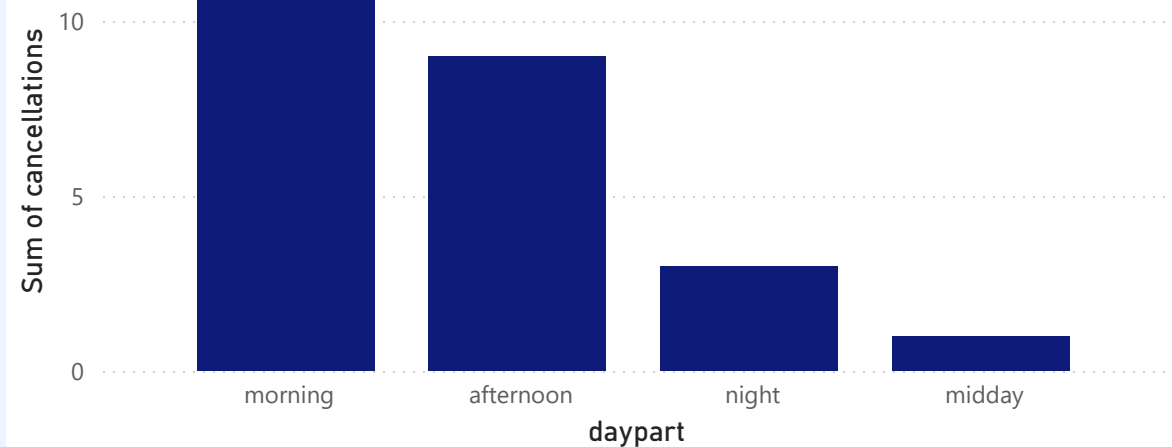
Q4: At what time of day (morning, midday, afternoon, night) do most reservation cancellations occur?

6.00

Average of cancellations

daypart	cancellations
afternoon	9
midday	1
morning	11
night	3

Number of cancellations per day

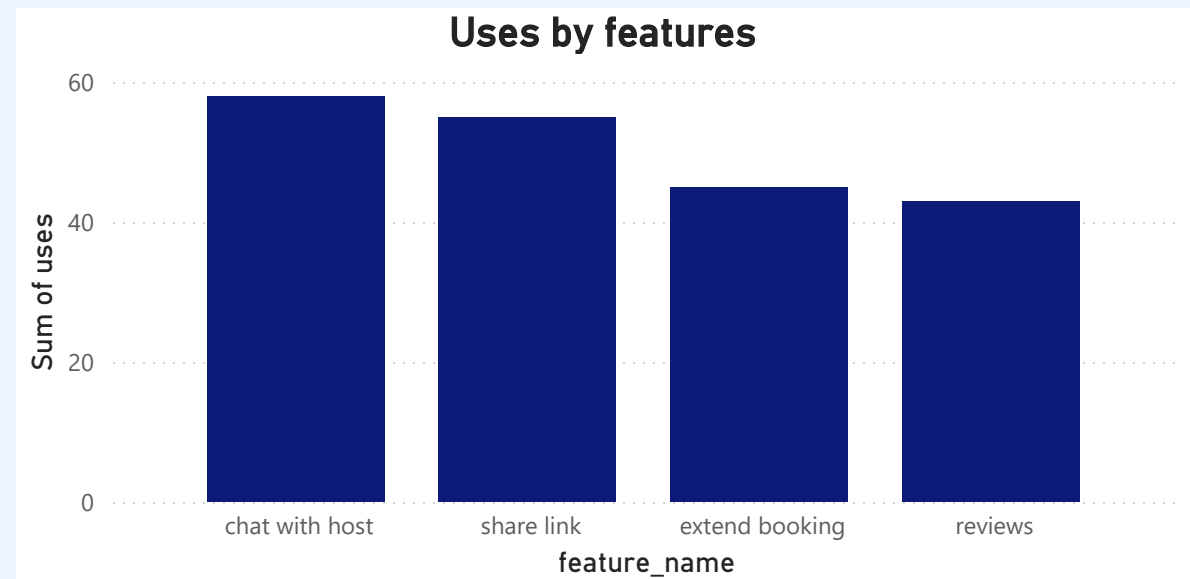


Q7: Which app features (extend booking, share link, reviews, chat with host) are the least used and should be removed or redesigned?

50.25

Average of uses

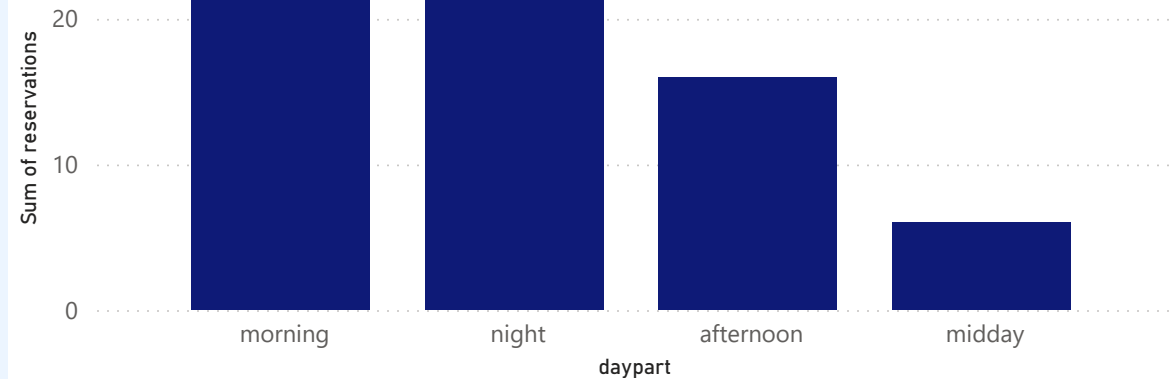
feature_name	Sum of uses
chat with host	58
extend booking	45
reviews	43
share link	55
Total	201



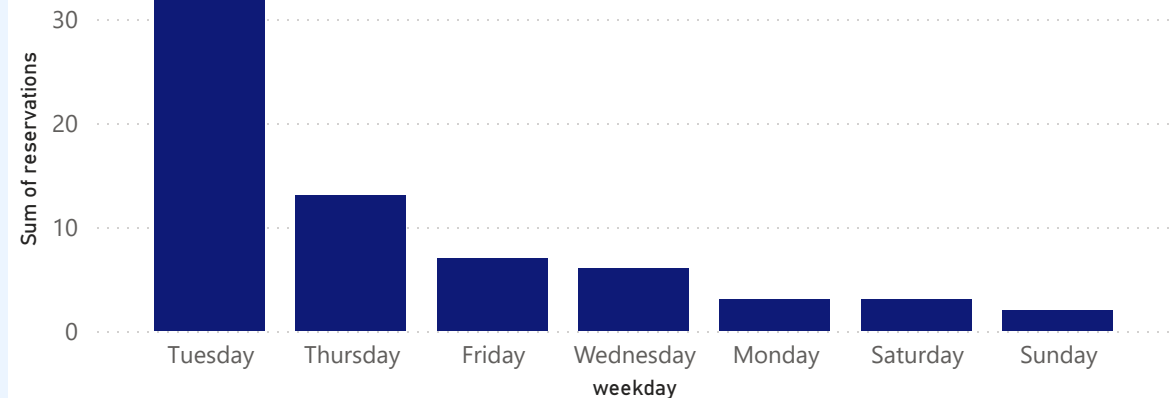
Q11: What patterns exist in space demand by day of the week and time slot (usage peaks)?

daypart	weekday	reservations
afternoon	Friday	1
afternoon	Monday	2
afternoon	Thursday	4
afternoon	Tuesday	7
afternoon	Wednesday	2
midday	Monday	1
midday	Thursday	1
midday	Tuesday	4
morning	Friday	1
morning	Thursday	4
morning	Tuesday	15
morning	Wednesday	3
night	Friday	5

Reservation per day part

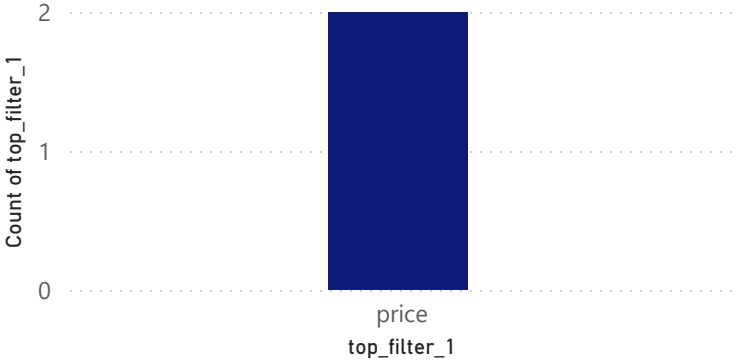


Reservation per week day



Q13: Which user segments (by major, semester, or age) book the most spaces, and what are their main preferences (booking telemetry + satisfaction surveys)?

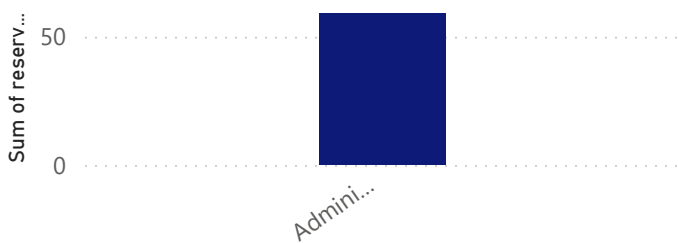
Count of top filters



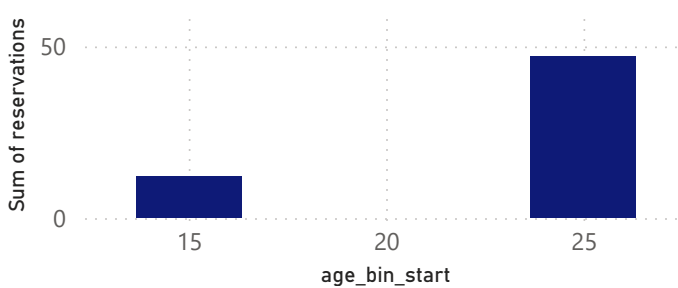
Reservations by student semester



Reservations by student majors



Reservations by student age



major	Sum of reservations	Average of semester	Average of avg_satisfaction	Average of age_bin_start
Administración	59	5.50	3.68	20.00
Total	59	5.50	3.68	20.00